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General Knowledge
World Geography

STUDY NOTES

GEOMORPHOLOGY: LANDFORMS(MARINE)

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GEOMORPHOLOGY: LANDFORMS(MARINE)

Marine Landforms and Cycle of Erosion

- **Sea waves, aided by winds, currents, tides and storms** carry on the erosional and depositional processes.
- **The erosive work of the sea depends upon size and strength of waves, slope, the height of the shore between low and high tides**, the shape of the coast, the composition of rocks, depth of water, human activity etc.
- **The wave pressure compresses the air trapped inside rock fissures, joints, faults, etc.** forcing it to expand and rupture the rocks along weak points. This is how rocks undergo weathering under wave action.
- **Waves also use rock debris as instruments of erosion (glaciers are quite good at this).** These rock fragments carried by waves themselves get worn down by striking against the coast or one another.
- The solvent or chemical action of waves is another mode of erosion, but it is pronounced only in case of soluble rocks like limestone and chalk.

Marine Erosional Landforms

Chasms

- These are narrow, deep indentations (a deep recess or notch on the edge or surface of something) carved due to headward erosion (downcutting) through vertical planes of weakness in the rocks by wave action.
- With time, further headward erosion is hindered by lateral erosion of the chasm mouth, which itself keeps widening till a bay is formed.

Wave-Cut Platform

- When the sea waves strike against a cliff, the cliff gets eroded (lateral erosion) gradually and retreats.
- The waves level out the shore region to carve out a horizontal plane or a wave-cut platform.
- The bottom of the cliff suffers the maximum intensive erosion by waves and, as a result, a notch appears at this position.

Sea Caves

- Differential erosion by sea waves through rock with varying resistance across its structure produces arched caves in rocks called sea caves.

Sea Arches

- When waves from opposite directions strike a narrow wall of rock, differential erosion of the rock leaves a **bridge-like structure** called Sea arch.

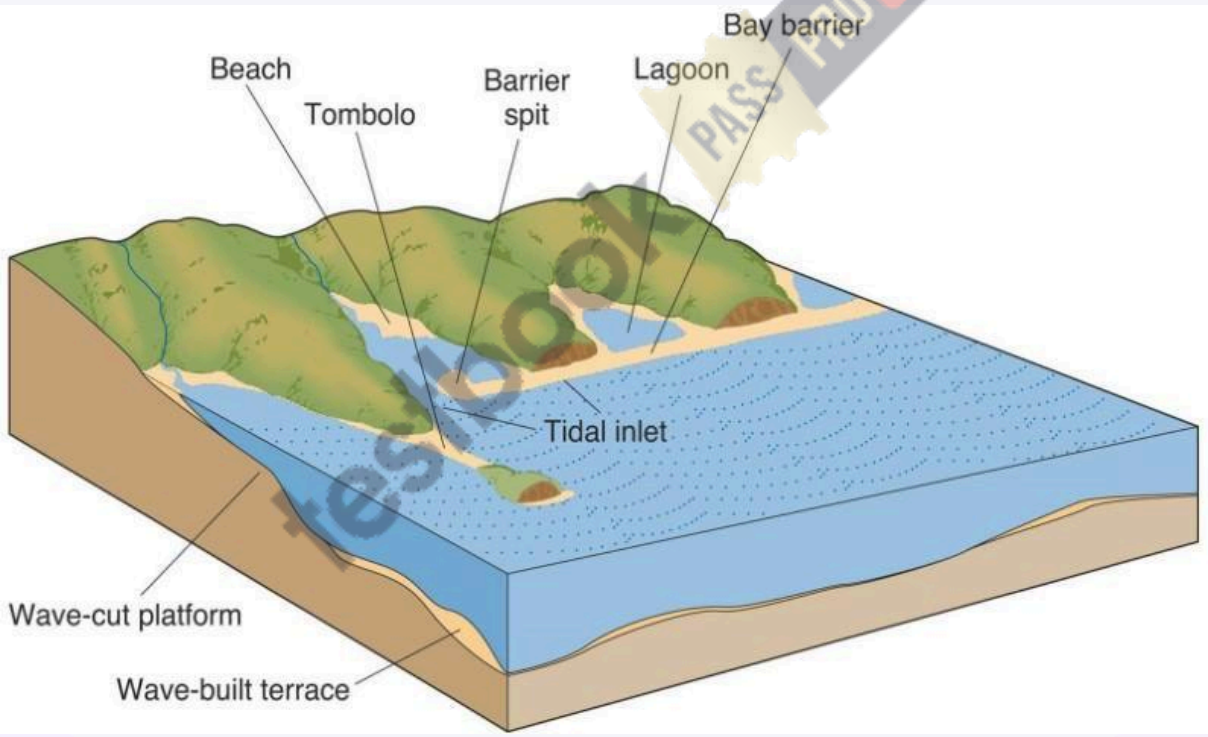
Stacks/Skarries/Chimney Rock

- When a portion of the sea arch collapses, the remaining column-like structure is called a stack, skarry or chimney rock.

Hanging Valleys

- If the fluvial erosion of a stream at the shore doesn't match the retreat of the sea, the rivers appear to be hanging over the sea. These river valleys are called hanging valleys.

Marine Depositional Landforms



Beach

- This is the temporary covering of rock debris on or along a wave-cut platform.

Bar

- Currents and tidal currents deposit rock debris and sand along the coast at a distance from the shoreline.

- The resultant landforms which remain submerged are called bars.
- The enclosed water body created is called a **lagoon.**

Barrier

- It is the overwater counterpart of a bar.

Spit and Hook

- A spit is a projected deposition joined at one end to the headland, with the other end free in the sea.
- The mode of formation is similar to a bar or barrier.
- A shorter spit with one end curved towards the land is called a **hook.**

Tombolos

- Sometimes, islands are connected to each other by a bar called tombolo.

Ria

- When streams dissect a region into a system of valleys and divides, submergence produces a highly irregular shoreline called ria coastline.
- The coast of south-west Ireland is a typical example of the ria coastline.

Fjord

- Some coastal regions have been heavily eroded by glacial action, and the valley glacier troughs have been excavated below sea level.
- After the glaciers have disappeared, a fjord coastline emerges.
- These coasts have long and narrow inlets with very steep sides.
- The fjord coasts of Norway are a typical example.